

# **Chris Controller**

Static Oxygen Concentrator Remote  
Control System



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## Safety Information:

The following information is important for the safe operation of the Chris Controller system. Read through the entire manual and familiarize yourself with the system before using the system. Failure to do so may result in injury or death. Pretty dramatic, huh! Now you have to ask yourself if you really want to use something that could result in injury or death.

**Rotating Machinery:** This system has exposed rotating gears. Do not wear dangling or loose fitting clothing or jewelry that may get caught in the rotating gears. The Controller uses the powerful 28BYJ-48 motor in the 12V version. While it may not be enough to harm a finger, it can do serious damage to the stoutest of commercial plastic drinking straws.

**Choking Hazard:** There is nothing edible in this system. Every effort was made so that it doesn't look like food. If you try to eat it, you will probably choke on it. If you're that hungry, just get a candy bar instead.

**Not a Toy:** Apparently some people think it's a good idea to give children plastic bags to play with. That is why so many manufacturers say not to play with plastic bags. To prevent you from giving children plastic bags as toys, this product is supplied without any plastic bags. Give your children a proper toy instead.

**Prop. 65 Warning:** Like everything else, this can cause cancer in the state of California. So do the safe thing and keep you and your loved ones out of the state of California!

**Psychological Hazard:** Unfortunately oxygen concentrator flow rates are typically metric (L/min). While no one knows what an "L/min" is (everyone knows what a "cubic foot per hour" is, though) this system uses metric so that it remains consistent with the oxygen concentrator system of units. For that reason, all parts of the design of this system is in metric units. For example, the set screws used are metric and use metric hex wrenches (2.5mm). While the metric system can cause psychological harm, it was determined the mixing of units would be more damaging. Thus only metric is used in the Chris Controller system.



## Introduction

Stationary oxygen concentrators are typically loud and cannot be placed in rooms where one is trying to sleep or reside. Long lengths of tubing are used to connect the user to the concentrator, so the user is not in the same room with the concentrator. This presents a problem when the user needs to change the oxygen flowrate. For example, if the resting oxygen flow rate is lower than the flow need to get up and walk, then the user must increase the oxygen flow before getting up. But if the concentrator is not in the same room, another person must be available to change the flowrate.

The technical paper from Duke University titled “Design and Development of a Novel System for Remote Control of Stationary Oxygen Concentrator Flow Rate” by authors Nikhil Gadiraju, Nikhil Peterson, Jessica Shah, Annabelle Chu, Michael A Larbie II, Amy Bu, Ann Saterbak was the basis for this project (See Attachment 1).

The Chris Controller system was designed to work on the DeVilbiss Drive 10 Liter Oxygen Concentrator. This concentrator uses a manual flow meter to adjust the delivered Oxygen flow. This is similar to many concentrator designs, as such; the Chris Controller is designed to be easily adaptable to other concentrator designs.

The Controller works by grabbing the knob that adjusts the oxygen flow and turning the knob by use of a stepper motor. The Controller system is delivered with a set of three gears to grab the knob. Each of these gears is labeled 1 through 3; choose the one that best fits your knob. Gear 1 has a straight opening and provides the greatest clearance for the knob. Gears 2 and 3 have a slight taper to the opening that fits the knob, with Gear 2 being a little tighter than Gear 3.

The remote control has a set of Up/Down buttons. When Up is pressed the gear on the knob will rotate counterclockwise. On the DeVilbiss concentrator, this increases flow. If your concentrator design is opposite, then the Up/Down buttons will work opposite (Up will decrease flow, Down will increase flow).

The intent of the system is that no modifications to the concentrator are necessary. Since many people do not own the concentrator, but are instead supplied via a medical supply company, modification of the concentrator was not an option. The knob gear grabs the flow knob with a set of 4 set screws. The Controller itself is attached to the concentrator with the use of double sided tape (e.g., 3M double sided tape). No drilling or screwing into the concentrator is required for the Controller system to function.

The system includes:

- Chris Controller
- Chris Controller Remote
- Set of 3 knob gears (only one required for operation)
- AC Adapter (12VDC Output)
- Four AAA alkaline batteries

The Controller is powered by the AC adapter. The Remote uses four AAA batteries. The life of the batteries is dependent upon usage of the system.

The brains of the system are the Espressif ESP32-S2-Solo-2 microcontroller modules. These modules use an 802.11 Wi-Fi standard for communication. But instead of requiring a connection to a Wi-Fi network, the Remote and Controller communicate using Espressif's ESP-NOW wireless protocol. This provides quicker response time than using a Wi-Fi network, while providing the superior range of Wi-Fi over Bluetooth. The Remote and Controller communicate on the 2.4GHz frequency (same as Wi-Fi networks). They use a "peer to peer" network based upon the MAC address of each ESP32 module. This information has been coded into the firmware; no pairing of devices is required by the user.

Documentation for this project, including source code and 3D models can be found at:

<https://github.com/tdevince/Chris-Controller>



## Chris Controller:

The Controller consists of a stepper motor, a multi-turn potentiometer, a set of gears (flow meter knob gear, drive gear and pot gear). The stepper motor is able to precisely move the flow meter knob (via the gearing) in either the up or down direction (counterclockwise or clockwise). The stepper motor does not have a stop and can continue to travel in one direction infinitely.

The stepper motor is not capable of knowing where in the flow range it currently resides. It only knows turn clockwise or counterclockwise by so many steps. The stepper motor used is the 28BYJ-48 motor. This motor is most commonly available in the 5V version; this Controller uses the 12V version. The motor itself has 32 steps per revolution, but is internally geared so that it takes a total of 2048 steps to complete one rotation of the output shaft.

The potentiometer is the memory of the system. A potentiometer is a variable resistor. By turning the potentiometer we can increase or decrease resistance. The potentiometer used in the Chris Controller is a 10 turn 10kOhm resistor (Bourns model 3590S-2-103L). This means at one end of travel the resistance is 0 ohms. At the other end of travel (10 turns later), the resistance is 10,000 ohms. One complete turn of the potentiometer is equal to 1000 Ohms. By including the potentiometer in the gear train, we turn the potentiometer the same time and the same amount as we turn the flow knob when the stepper motor is engaged. All gears have the same number of teeth, so every gear turns in synchronization with each other.

After calibrating the system, the resistance of the potentiometer is directly proportional to flow rate. If power is lost, the potentiometer will retain the flow setting of the concentrator.

The system will not allow the stepper motor to hit the hard stops of the potentiometer. To help keep the full range of the full knob within the operating range of the potentiometer, it is recommended that the lowest flow rate (2.0L/min) be set with the potentiometer approximately 1.5-2 turns up from the low range stop.

**Caution: Do not manually turn the flow knob, or gears while the drive gear is on the stepper motor. This may damage the internal gearing of the stepper motor.**

To set the starting location of the potentiometer, remove the driver gear from the stepper motor. The gears or a snug fit on both the potentiometer and stepper motor. The gears are held to the shaft of these components with a pair of set screws. The set screws use a 2.5mm hex wrench (Allen wrench). With the drive gear removed, rotate the potentiometer gear all the way clockwise (as observed from the top of the Controller) until the stop is hit. Then rotate the gear 1.5-2.0 turns clockwise. Now reinstall the drive gear. When placing Controller on the concentrator, set the flow rate to the lowest setting (2.0 L/min). This will prevent the lowest setting of the Controller from hitting the hard stop of the potentiometer. This will provide approximately 8.5 turns of operation of the system. The concentrator flow meter will typically go from 2.0 L/min to 10.0 L/min in less than 5 turns.

The Controller is powered from the provided 12V AC adapter. This AC adapter provides an output of 12VDC with a maximum output current of 1.0Amps. This is well below the current draw of the Controller. Do not power the Controller with more than 12VDC since the stepper motor is only rated for a maximum of 12VDC.

The Controller has two modes of operation:

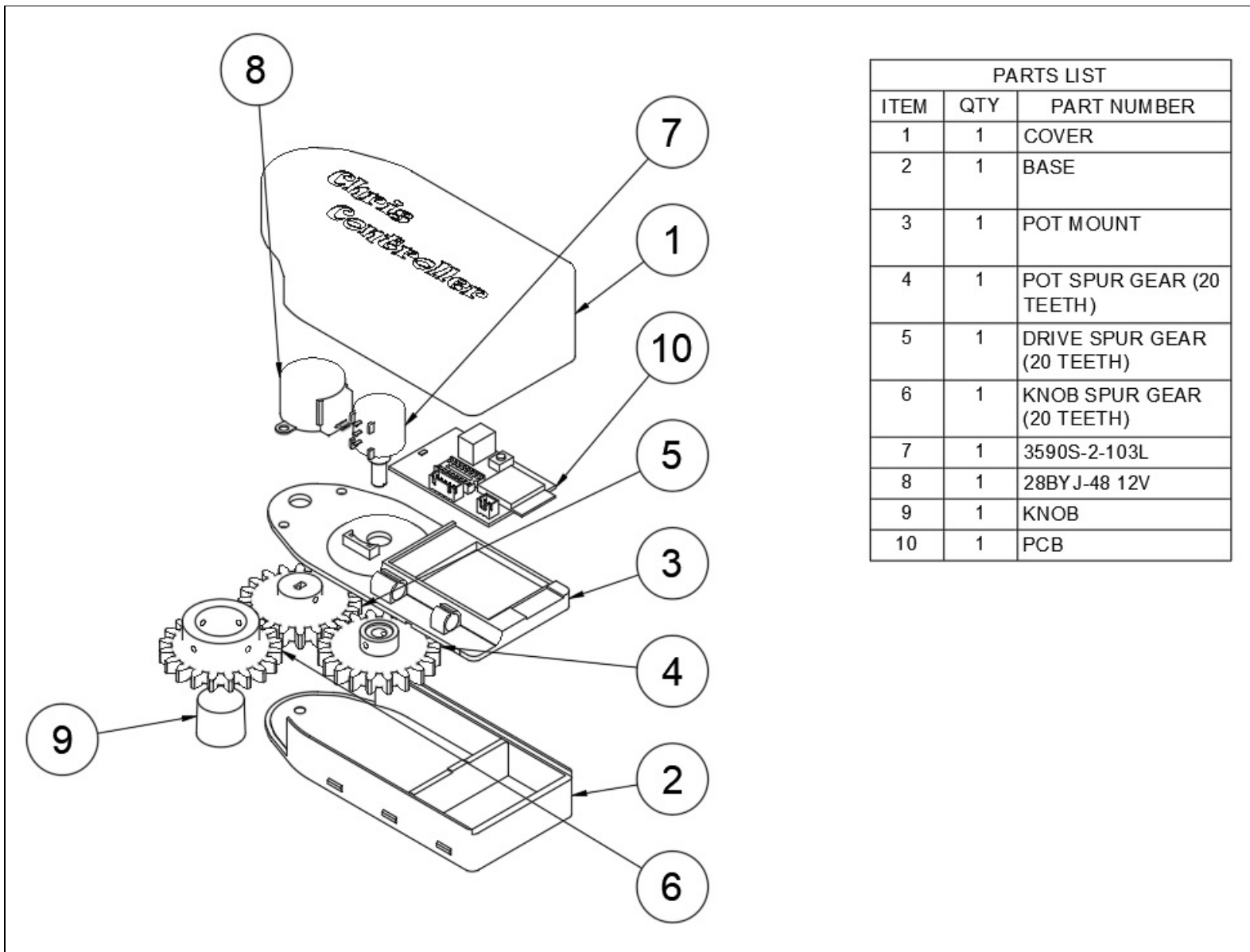
- Normal Mode
- Over-The-Air (OTA) mode

Operations to control the flow rate of the concentrator using the stepper motor occurs under the Normal Mode. Firmware updates to the Controller are only possible in the OTA mode. Firmware updates require a connection to a Wi-Fi network. The Printed Circuit Board (PCB) of the Controller has a button that will toggle the mode of the Controller from Normal Mode to OTA Mode. This button is only accessible by removing the top cover of the Controller to expose the PCB.

By pressing the OTA button on the PCB, the Controller will reboot in the firmware update mode. The Controller will store the SSID and Password to connect to the local network. If the Controller has not connected to this network previously, it will present a Wi-Fi Manager Webpage that may be used to store the network SSID and Password.

While connected to a network, firmware updates may be carried out from a PC connected to the same network. Once the update is completed, pressing the OTA button on the PCB again will return the Controller to Normal Mode.

See the section on firmware updates for details.



**Figure 1: Controller Exploded Diagram**



## Remote:

The remote control is the user interface for the Chris Controller system. The Remote has two buttons, Up and Down, that allow the user to command the Controller to rotate the flow knob counterclockwise and clockwise, respectively. For concentrators where counterclockwise increases flow, the Up and Down button will increase and decrease flow, respectively.

The Remote also features a 128x64 OLED display screen. This screen will show the current oxygen flow rate for which the Controller is set. The upper right part of the display also provides an icon reflecting the remaining capacity of the 4 AAA alkaline batteries.

The upper left of the Remote has access to the Calibration feature of the Remote by inserting a paper clip (or similar object) into the hole marked “Cal”. This will press the “Cal” button on the PCB. Once the “Cal” button is pressed, instructions on how to calibrate the system is provided.

While the potentiometer of the Controller stores the current position of the concentrator flow knob, the Remote is the device that correlates the potentiometer resistance to an actual flow rate. The calibration information is stored in non-volatile memory; therefore the information is retained after the Remote has been powered down.

A slide switch on the side of the Remote turns the power on/off from the batteries to the PCB. This slide switch physically interrupts the battery to PCB connection. While powered up, to conserve batteries, the Remote display and Remote ESP32 module will go to sleep after 15 seconds of inactivity (while in the normal mode of operation). To wake the Remote, a single press of either the Up or Down button will wake the Remote. If the system is not going to be used for an extended period of time, the slide switch is the preferred method to turn the Remote off since this will conserve battery life.

The Remote during normal operation consumes approximately 60-70mA of current. When in sleep mode, this value drops to approximately 0.14mA. A typical AAA battery provides approximately 1000mA-hr of capacity. If the Remote were to be left in sleep mode, the batteries should last approximately 300 days. This value will vary by the frequency of use and the quality of batteries used. The slide switch eliminates all “vampire” loads, so that 0.0 mA are used from the batteries while powered off.

Similar to the Controller, the Remote has an OTA button on the PCB. This button is only accessible when the PCB is exposed.

The top of the Remote has a post and hole that is designed to have a lanyard attached. This feature will help avoid the dreaded “Where did I put that @\*%&@#?ing Remote” crisis.

The Remote has 3 modes of operation:

- Normal Mode
- Calibration Mode
- OTA mode

### ***Normal Mode:***

Normal Mode is used for controlling the oxygen flow via the Controller. By holding the Up or Down button, the display will increase the flow demand in 0.5 L/min increments. 2.0L/min being the lowest and 10.0 L/min being the maximum flow settings possible. Once the desired flow rate is displayed on the Remote, release the buttons. When the buttons have been released for 1 second, the Remote will send the new flow demand to the Controller. The Remote will display the message “waiting...” while the Controller adjusts the flow rate. When the Controller reaches the desired flow rate, it will respond to the Remote with the current potentiometer setting. The Remote will convert this value and display it as the new oxygen flow rate.

After 15 seconds of inactivity, the Remote will go to sleep. Pressing either the Up or Down buttons will wake the Remote. Upon waking the Remote will inquire from the Controller its current position. When the Controller responds with the current potentiometer setting, the Remote will convert this value and display the corresponding oxygen flow rate.

### ***Calibration Mode:***

The Remote must be calibrated to be able to convert potentiometer values to an oxygen flow rate. Calibration is only required when initially setting up the Chris Controller. Once it has been set up, the calibration data is retained in non-volatile memory. The only time a Calibration would be required is if the flow knob is moved independent of the Controller (i.e. the flow knob on the concentrator manually moved with the Controller not installed). If the Controller is ever removed and reset on the concentrator, the calibration procedure should be repeated.

**Caution: Do not manually turn the flow knob, or gears while the drive gear is on the stepper motor. This may damage the internal gearing of the stepper motor.**

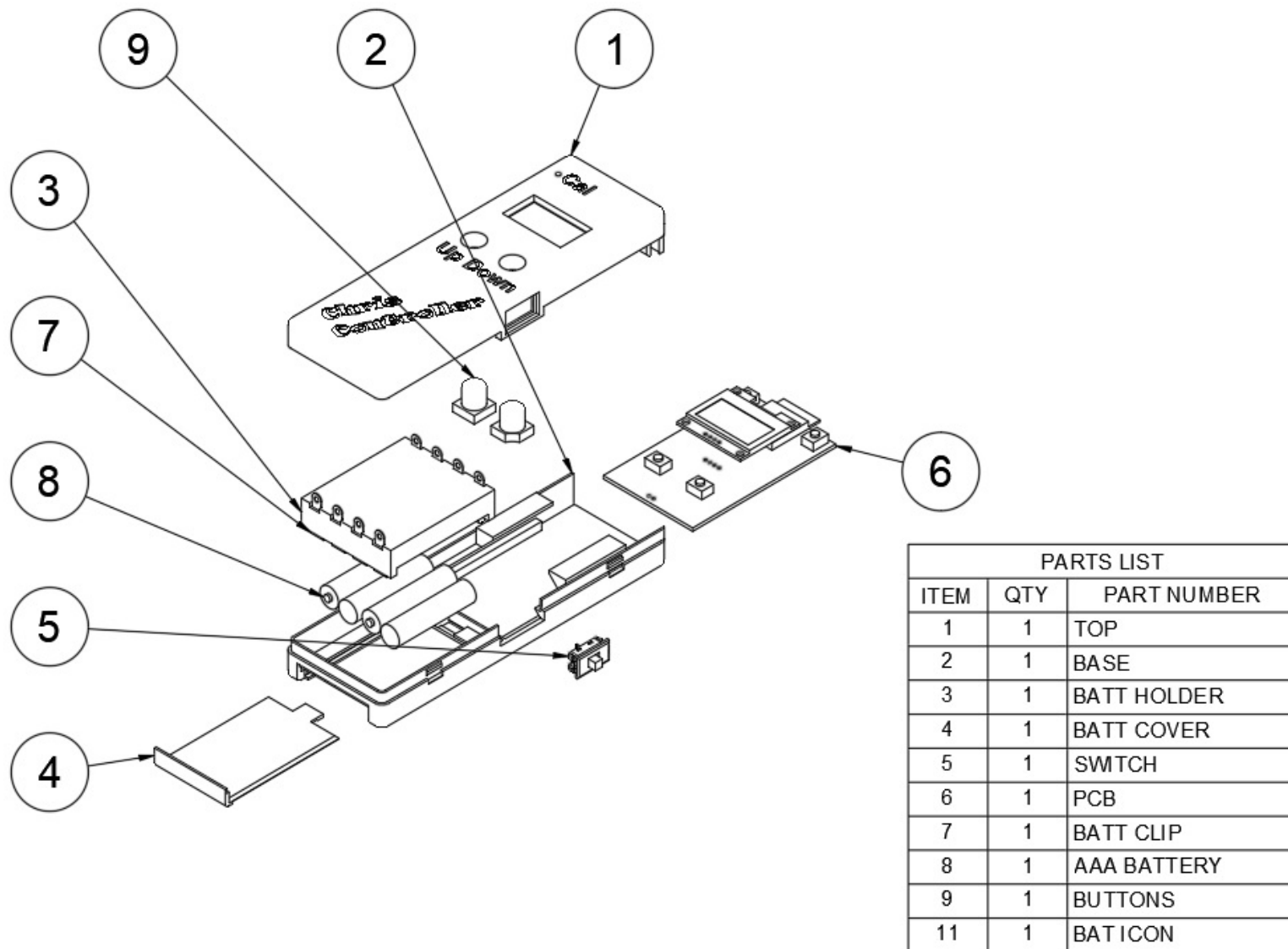
### ***OTA Mode:***

The OTA mode allows firmware updates of the ESP32 module of the Remote. While in Normal or Calibration Mode, the Remote is using the ESP-NOW protocol to communicate with the Controller. To enable firmware updates, the Remote must disconnect from the Controller and connect to the local Wi-Fi network.

Pressing the OTA button on the PCB will toggle the OTA Mode, and reboot the ESP32 module. Upon booting up, the ESP32 module will log on to the local Wi-Fi network using its stored SSID and Password for the Wi-Fi network. If the Remote has not connected to the Wi-Fi network previously, it will deploy its Wi-Fi Manager Webpage. This Webpage may then be used to log onto the network. Once logged on to the Wi-Fi network, the firmware on the ESP32 module may be performed with a PC connected to the same Wi-Fi network.

When logged onto the local network in OTA mode, the Remote display screen will show that the Remote is in OTA update mode and will also display its IP address. While in OTA mode, the Remote will not go to sleep.

After updating the firmware, another press of the OTA button will reboot the Remote in the Normal Mode. See the section on firmware updates for details



**Figure 2: Remote Exploded Diagram**





## Installation and Setup:

The Chris Controller only requires 8 simple steps for installation. The required items for this installation that were not provided with the system are:

- 2.5mm hex wrench (Allen wrench)
- Double sided tape

### Step 1:

Determine where on the concentrator the Controller should be located. On the DeVilbiss Drive 10 Liter Oxygen Concentrator, the Controller is designed to be affixed to the right of the flowmeter and flow knob. The Controller should not be installed such that the flowmeter is blocked. The drive and flow knob gears are designed to have a lateral separation of 48mm (1.9"). So the Controller should be placed such that the drive gear is offset this distance from the flowmeter.

### Step 2:

Choose the appropriate flow knob gear to use. The system is supplied with three gears, each labeled 1, 2 and 3. Gear 1 has a straight wall opening to accommodate the knob, and will provide the greatest clearance. Gears 2 and 3 have tapered openings to accommodate the knob, with Gear 2 having tighter tolerances than Gear 3. The knob to choose will be the one that fits loosely around the knob but minimizes the clearance between the knob and gear.

Attach the knob by gradually tightening the 4 set screws. These screws should be used to align the gear so the knob is as close to the center of the opening as possible. By maintaining concentricity between the knob and gear, binding of gears can be minimized. Do not overtighten the set screws. It is only necessary to snug the screws so they adequately grab the knob and center the gear.

### Step 3:

Dry fit the location of the Controller so that the drive gear meshes with the knob gear installed in Step 2 above. The mesh should be slightly loose to prevent gear binding, but not so loose as to have significant backlash in the gearing. It may be necessary to reposition the knob gear up or down to ensure the drive gear and knob gear are at approximately the same height. Bear in mind when dry fitting that using a foam style double sided tape will slightly elevate the drive gear.

### Step 4:

The Controller was delivered with the potentiometer set to 1.5 to 2.0 turns from the bottom stop. Therefore set the flow knob on the concentrator close to 2.0L/min to provide adequate margin when calibrating.

### Step 5:

Place double sided tape on the bottom of the base of the Controller, and stick it to the concentrator such that the knob gear and drive gear mesh as described in Step 3.

Step 6:

Plug in the AC adapter and then plug it into the Controller to power the Controller.

Step 7:

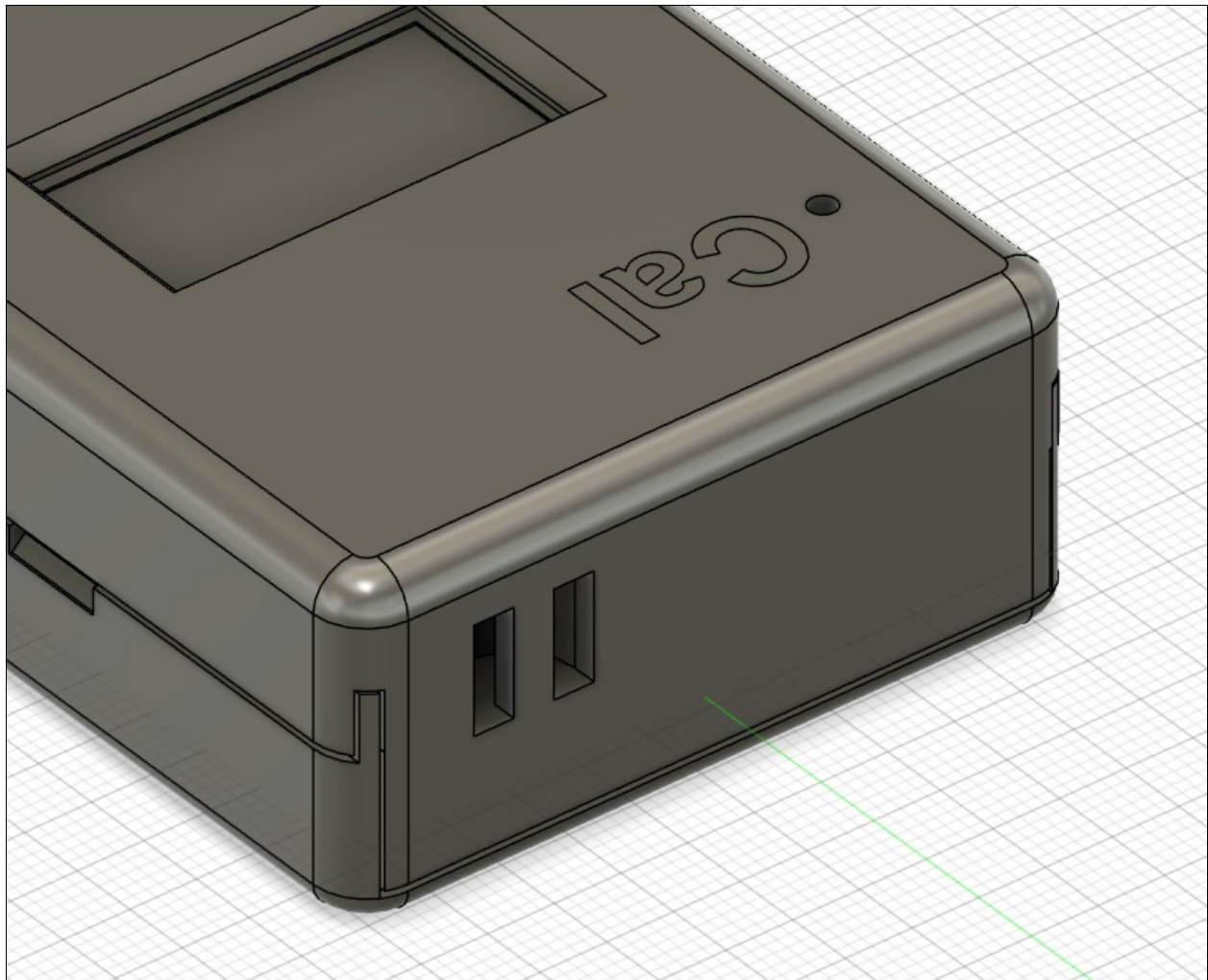
Install 4 AAA batteries in the Remote, and then switch the power on the Remote. It will provide a Controller reading, this value is not accurate until the Remote has been calibrated.

Step 8:

Calibrate the Remote (See Calibration Section).

After completion of the calibration step, the Chris Controller system is ready for operation.

An optional lanyard may be attached to the post in the top of the Remote.



**Figure 3: Remote Lanyard Post and Hole**

## Calibration:

Calibration of the system should be done anytime the knob and potentiometer gear are out of synch with each other. This will, at a minimum, be at the first installation of the system. Any other time when the knob is manually moved independent of the gear train, a calibration should be performed.

**Caution: Do not manually turn the flow knob, or gears while the drive gear is on the stepper motor. This may damage the internal gearing of the stepper motor.**

Calibration should be performed at the oxygen concentrator to allow visual readings of the flow rate during the process. The calibration process involves using the Remote to command the stepper motor to increase or decrease flow by pressing the Up or Down buttons. The Remote will identify the target flow rate for each step. When the target flowrate is displayed by the Remote, use the Up/Down buttons until the flowrate displayed on the oxygen concentrator matches the target value given by the Remote. When the target flowrate is achieved, pressing the Up and Down buttons simultaneously will enter this value into the Remote.

The stepper motor uses 2048 steps per revolution. Using the Remote Up or Down buttons will cause the stepper to in increments of 128 steps. This provides a resolution of 16 steps per revolution, or 22.5°. This should provide adequate resolution for level of accuracy of the concentrator flowmeter.

The Remote will ask to repeat this process for a total of 9 data points from 2.0 to 10.0 L/min in 1.0L/min increments. After the last data point has been entered, the calibration data are saved to memory and the Remote will return to the Normal Mode.

To Calibrate the system, perform the following steps (note, the Remote will not go to sleep during calibration):

### Step 1:

Ensure the Controller is powered up, turn on the Remote and wait for the display to show the current Controller position.

### Step 2:

If the Remote is asleep, first wake the remote by pressing either the UP or Down button. Using a paper clip, or other similar object, insert it into the “Cal” hole located on the Remote. Press the button on the PCB with the paper clip. The Remote will now enter calibration mode and display a screen of instructions:

Cal. Mode: Use Up or  
Dwn to turn controller  
To desired flow. Press  
Up & Down to ‘Enter’

Press the Up and Down button to progress to the next page.

**Step 3:**

After pressing the Up and Down buttons in Step 2, the user will be presented with the first target flow rate. Using the Up/Down buttons have the Controller increase/decrease flow on the concentrator until the flowmeter on the concentrator matches the target value displayed on the Remote. When the target value is achieved, simultaneously press the Up and Down buttons to enter the value. The Remote will then prompt the user for the next data point.

**Step 4:**

Repeat Step 3 until all data points from 2.0 through 10.0 L/min have been entered. The Remote only uses cardinal values for calibration. For values between the cardinal values (e.g., 2.5, 3.5... etc.), the Remote will linearly interpolate between the cardinal values stored.

After the target value for 10.0L/min has been reached and the Up and Down buttons pressed, the Remote will save the calibration data and return the Remote to Normal Mode.

If the Cal button is pressed again before the calibration procedure is completed, the Remote will return to Normal Mode without saving the calibration data. Similarly, if the Remote is switched off before the calibration procedure is completed; the Remote will return the Normal Mode when powered on again without saving the calibration data.

## Firmware Updates:

The firmware on the Controller as well as the Remote can be updated “Over-the-Air” (OTA) using a local Wi-Fi network. During Normal and Calibration modes of operation, the Controller and Remote communicate without connecting to a Wi-Fi network. This simplifies normal operation and significantly increases system response time.

However to perform a firmware update, the unit being updated must disconnect from this “peer to peer” network and connect to a local Wi-Fi network. Both the Remote and the Controller have an OTA button the PCB that is not accessible when the top cover is installed. If the Remote is asleep, it will be necessary to wake the Remote by pressing the Up or Down button first.

Pressing the OTA button will reboot the Controller/Remote and look for the local Wi-Fi network for which it has stored SSID and Password information. If this is the first time connecting to the network, the Controller/Remote will start up its Wi-Fi manager Webpage. To access this Webpage, go to Settings>Wi-Fi on your iPhone (Android devices may be different). Then connect to the Wi-Fi manager shown in the list of Wi-Fi networks. For the Controller this will be labeled “Chris\_Controller”, “Chris Remote” for the Remote.

The Wi-Fi Manager Webpage will then be displayed; the first option is to configure the Wi-Fi connection. Selecting this option will open a page requesting the network SSID and password. Scroll down through the list of displayed networks until you find the desired network. Selecting that network will pre-fill the SSID information. Next enter the password for that network and connect to the network.

If the SSID and Password are properly filled in, the Controller/Remote will reboot and connect to that network. If unsuccessful, the Controller/Remote will relaunch the Wi-Fi manager. It will be necessary to reconnect with your iPhone and re-enter SSID/Password. Typically connection errors occur when the Wi-Fi signal is weak, or a something was mistyped for SSID/Password.

Once connected to the Wi-Fi network, the Remote will display the IP address that it is using to connect to the network. The Controller has no such display, to find the IP address it will be necessary to find it from the network. Most Wi-Fi routers allow the user to see devices connected to the network. The device name for the Controller should show up in that list as “Chris\_Controller”.

The firmware on both the Remote and Controller was programmed using Microsoft’s Visual Studio Code (VS Code) with the PlatformIO plug-in. To update the firmware using PlatformIO, use the espota protocol and specify the device IP address in the platformio.ini file. For example, for an IP address of 192.168.0.144, the following lines would be added to the platformio.ini file:

```
[env:esp32-s2-saola-1-ota]
upload_protocol = espota
upload_port = 192.168.0.140
```

The status of the upload will be displayed in the terminal window of VS Code. Once the update is shown to be completed and successful, pressing the OTA button on the Remote/Controller will reboot the device and start in Normal Mode with the updated firmware.

## **Attachment 1:**

### **Design and Development of a Novel System for Remote Control of Stationary Oxygen Concentrator Flow Rate**

by Nikhil Gadiraju, Nikhil Peterson, Jessica Shah, Annabelle Chu, Michael A Larbie II, Amy Bu,  
Ann Saterbak





# Design and Development of a Novel System for Remote Control of Stationary Oxygen Concentrator Flow Rate

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**Purpose:** Long-term oxygen therapy involves utilizing stationary oxygen concentrators to allow patients with respiratory illnesses to attain sufficient blood oxygenation via supplemental oxygen. Disadvantages of these devices include their lack of remote adjustability and domiciliary accessibility. To adjust oxygen flow, patients typically walk across their homes – a physically taxing activity – to manually rotate the knob of the concentrator flowmeter. The purpose of this investigation was to develop a control system device that allows patients to remotely adjust the oxygen flow rates on their stationary oxygen concentrator.

**Methods:** The engineering design process was used to develop the novel FLO2 device. The two-part system is composed of 1) a smartphone application and 2) an adjustable concentrator attachment unit that mechanically interfaces with the stationary oxygen concentrator flowmeter.

**Results:** Product testing indicates that users successfully communicated to the concentrator attachment from a maximum distance of 41m in an open field, suggesting usability from anywhere inside a standard home. The calibration algorithm adjusted oxygen flow rates with an accuracy of  $\pm 0.019$  LPM and a precision of  $\pm 0.042$  LPM.

**Conclusion:** Initial design testing suggests the device as a reliable and accurate method of wirelessly adjusting oxygen flow on a stationary oxygen concentrator, but further testing should be performed on different stationary oxygen concentrator models.

**Keywords:** oxygen therapy, respiratory disease, flow meter, wireless adjustment, medical device

## Introduction

More than 1.5 million patients in the United States utilize supplemental oxygen to manage various respiratory conditions, including chronic obstructive pulmonary disease, lung cancer, cystic fibrosis, and amyotrophic lateral sclerosis.<sup>1</sup> Patients with these respiratory diseases are unable to perform the normal physiological exchange of oxygen and carbon dioxide, resulting in oxygen deficiencies or hypoxia. Although the lung damage and resulting deficiencies associated with chronic obstructive pulmonary disease (COPD) are irreversible, home oxygen therapy can increase life expectancy and standard of living for patients who would otherwise struggle to complete daily tasks.<sup>2</sup> More specifically, COPD patients experiencing chronic hypoxemia have demonstrated susceptibility to the health advantages and quality of life improvements afforded by long-term home oxygen therapy.<sup>3–5</sup>

Advancements in home oxygen delivery have enabled both stationary and portable oxygen concentrators to provide supplemental oxygen. Portable oxygen concentrators are utilized primarily in mobile, lightweight applications, and in the absence of a permanent power source.<sup>6</sup> Most home oxygen therapy patients that are prescribed long-term oxygen therapy receive supplemental oxygen for over 15 hours a day through a stationary oxygen concentrator.<sup>3,7</sup> Oxygen concentrators function using a technique called pressure swing absorption in which nitrogen from atmospheric air is adsorbed onto

zeolite, a porous aluminosilicate material, producing an oxygen-rich gas. With this method, the oxygen concentration of 21% in atmospheric air can be increased to 90–95% in the gaseous product.<sup>8</sup> Patients require different amounts of supplemental oxygen when active, resting, and sleeping. Excessive oxygen can result in hyperoxia, while insufficient oxygen can result in hypoxia. Therefore, accurate adjustment of oxygen flow rate is necessary to avoid hypoxia and hyperoxia throughout daily changing activities.<sup>2</sup> Oxygen supplementation may need to be increased during periods of increased demand when patient activity increases, such as during exercise or when walking.<sup>9</sup> Prior studies have shown that patients with COPD and moderate hypoxia frequently have significant periods of poor oxygenation during daily activities and at night.<sup>10</sup>

As currently designed, patients or caregivers need to manually adjust the knob on the front of the device to increase or decrease oxygen flow rates. As a result, patients must walk across their homes – and sometimes up and down stairs – to turn the knob of the stationary oxygen concentrator flow meter to maintain adequate oxygenation. However, manual adjustment of oxygen flow is time-consuming and burdens patients, which often reduces therapy compliance.<sup>11</sup> Patients undergoing long-term oxygen therapy have previously expressed a need for remote control capabilities for stationary oxygen concentrators.<sup>12</sup> Furthermore, the American Thoracic Society (ATS) has highlighted the development of remote-control flowrate adjustment technology as a key funding priority for overcoming barriers to supplemental oxygen delivery via stationary oxygen concentrators.<sup>1</sup>

The limitations of current oxygen delivery methods led to the advances of new intelligent portable oxygen concentrators. For example, Sanchez-Morillo et al developed a portable oxygen concentrator that uses physiological sensors and predictors of oxygen desaturation to automatically adjust oxygen flow rates.<sup>13</sup> However, there are few developments in stationary concentrator technology that address the need for improved oxygen flowrate control and overall device accessibility.<sup>1</sup>

To adapt oxygen therapy to the dynamic needs of patients and minimize oxygen desaturation episodes, we developed FLO2, a novel oxygen concentrator control system that is an external add-on for existing stationary concentrators. FLO2 allows patients to wirelessly control the flow rate of their oxygen concentrator using a mobile application. This innovation removes any physical activity associated with oxygen flow adjustment for both patients and family members, which will lower the risks of hypoxia and hyperoxia associated with home oxygen therapy.

## Materials and Methods

### Engineering Design Process

To design and build the FLO2 oxygen concentrator control system, the team used the engineering design process.<sup>14</sup> The engineering design process is an iterative workflow utilized to create novel products. As a part of this process, a problem was identified and researched, potential solutions were brainstormed and evaluated using a set of design criteria, a prototype solution was designed and produced, and this initial prototype was iteratively improved upon following testing. In narrowing down the initial set of potential solutions, the design criteria that were utilized during solution evaluation were: range of 30 meters, mobile application reliability of 97.5%, flow rate accuracy of 0.25%, and flow rate precision of 0.25% (Table 1). In summary, the team utilized the iterative engineering design process to produce FLO2, a device that met all established design criteria.

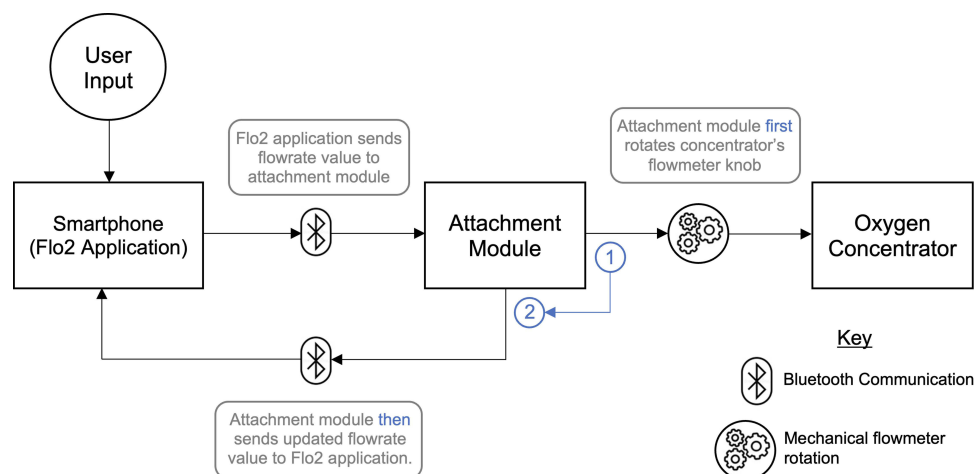
**Table 1** Design Criteria with Definitions

| Design Criteria     | Definition                                                                                        |
|---------------------|---------------------------------------------------------------------------------------------------|
| Bluetooth Range     | Maximum distance the app remains connected to the concentrator attachment                         |
| Reliability         | Bluetooth: Connects to attachment in under 10 seconds                                             |
|                     | App Adjustment: User action consistently register on the app and are communicated to attachment   |
| Flow Rate Accuracy  | Closeness of the flow rate observed on the concentrator compared to the expected target flow rate |
| Flow Rate Precision | Closeness of the flow rates observed on the concentrator for each target flow rate                |

Initially, a preliminary proof of concept was designed for both the communication and attachment modules. The initial communication module was composed of a simple Arduino-based Bluetooth transceiver that allowed users to utilize push buttons to elicit a verifiable response on another Arduino board; this proved user interface and wireless communication capabilities. The attachment proof of concept was a simple elastic clamping mechanism that modeled the module mounting contraption; all attachment proof of concepts were initially designed around the in-house oxygen concentrator, the Invacare Mobilaire V. Given these proof of concepts, iterative prototyping, involving rapid 3D-printing and design review cycles, was utilized to create an initial product prototype. This allowed for prototype iterations to be rapidly tested and refined until a final, viable prototype was created. This iterative prototyping process was repeated multiple times, with in-depth design reviews being conducted between processes. This led the product to transition from an attachment module designed purely for our in-house oxygen concentrator and a physical remote, to an adaptable attachment module and a mobile application.

For more informed refinements of the initial prototype, informal feedback was sought from community members with chronic lung diseases using home oxygen therapy (see [Appendix](#) for survey). From a total of 12 survey responses, 91% of responders were prescribed multiple oxygen flow rates each day depending on their level of exertion. Within this group, two members stated that they were prescribed certain flow rates for different levels of exertion but never changed their flow rate due to the difficulty in adjusting it. The survey received mixed feedback on the preference for a slider or button form of flow rate adjustment, so both methods were included in the development of the mobile application. Responders pointed to the ability to adjust to half increments using a slider and the ease of use of pressing a button when explaining their preferences. Finally, the community members expressed a high degree of interest in the continued development of this product.

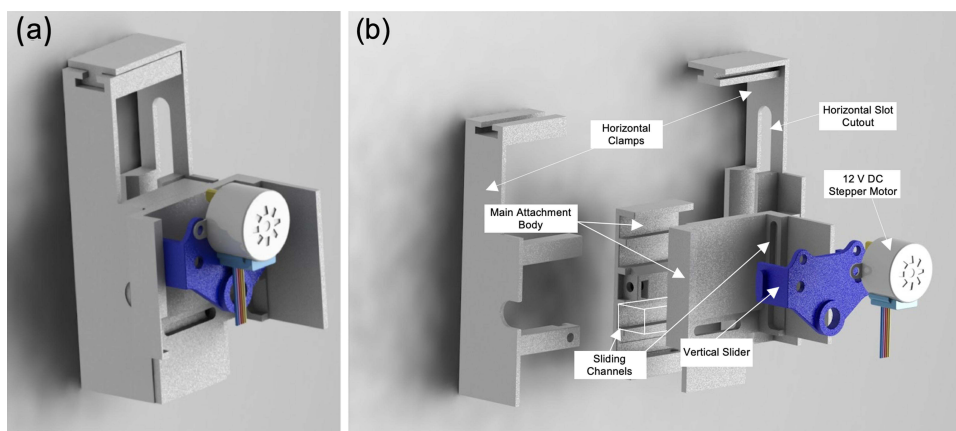
The final device solution, named FLO2, consists of two parts: 1) an adjustable oxygen concentrator attachment that clamps onto the flow meter of various oxygen concentrator models and 2) a mobile application that allows patients to wirelessly adjust their flow rate ([Figure 1](#)). The oxygen concentrator attachment is an external add-on that modifies the flow rate without interfering with the internal workings of the existing oxygen concentrator. The innovation allows patients to use an app on their smart device to change their flow rate from anywhere in their homes. After the FLO2 smartphone application receives user input, it wirelessly communicates with the oxygen concentrator attachment module via Bluetooth. This input prompts the attachment module to utilize a motor and gear system to mechanically rotate the knob on the flow meter of the oxygen concentrator accordingly. Subsequently, the attachment electronically updates the app with the new flow rate value via Bluetooth, which is then displayed to the user.



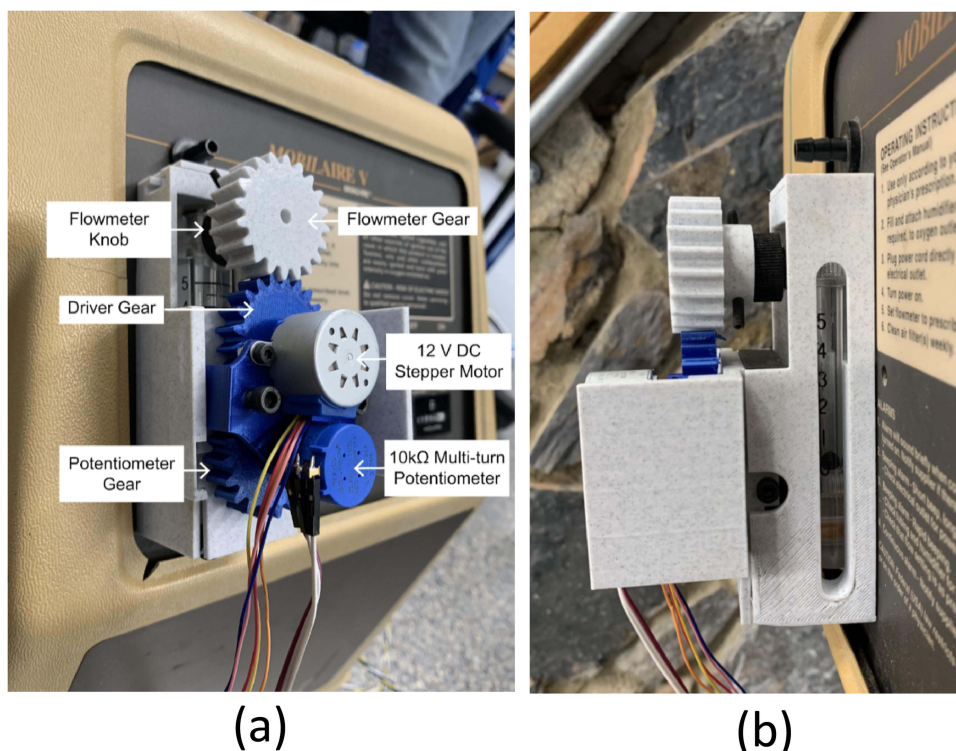
**Figure 1** Block diagram overview of FLO2 System. User input is sent to FLO2 application, which then prompts attachment to module to first (1) rotate flow meter knob and then (2) send updated flow rate value back to application.

## Attachment Module Housing

The attachment module was designed using Computer Aided Design (CAD) software and 3D-printed using polylactic acid (PLA). The attachment is set atop the oxygen concentrator flowmeter and works by using an adjustable horizontal clamping mechanism that laterally surrounds the flow meter (Figure 2). The right horizontal clamp was designed such that the user can read the oxygen flow rate ball position through the horizontal slot cutout in the case of product failure. Additionally, the vertical slider (Figure 2) can be appropriately adjusted to interface with the flow meter gear to allow for flow rate adjustment. The flow meter gear (Figure 3) attaches to the knob of the flowmeter using four adjustable set screws. This horizontal and vertical clamping mechanism was intentionally designed to allow for oxygen concentrator



**Figure 2** (a) CAD model of attachment module; (b) Exploded view of attachment module CAD model. The horizontal clamps and vertical slider function by moving through the appropriate sliding channels; they are held in place by tightening their respective set screws (not shown).



**Figure 3** (a) Concentrator attachment module and gear system clamped to flow meter; (b) Concentrator attachment side view with an insert designed for users to read the oxygen flow rate ball position.



adaptability. Rather than utilizing a static flowmeter slot for mounting, this designed clamp can adjust to various flowmeter and concentrator geometries, which allows this device to be compatible with many oxygen concentrator models.

## Motor and Potentiometer

The stepper motor was placed to rotate the motor driver gear, which was connected to a gear attached to the flow meter knob and a gear attached to the potentiometer (Figure 3). Electronically, a circuit was designed using a Bluno Beetle BLE microcontroller (DFR0339; DFRobot), a 10k $\Omega$  2W wire-wound linear multi-turn potentiometer (3590S-1-103L, Bourns Inc.), and a 12V 4 phase geared stepper motor (28BYJ-48; CenryKay). The microcontroller was powered using a 9V 2A AC/DC power supply (B087ZWS7CG; GuanTing), which was compatible with standard 110 V wall outlets; its power line runs alongside the oxygen concentrator's power line (Figure 4).

The microcontroller was programmed using Arduino to control the rotational direction and position of the stepper motor based on the potentiometer input. The potentiometer was used to track the position of the flow meter, even when the concentrator attachment or the oxygen concentrator is not on. This was implemented as a safety mechanism for the FLO2 control system to track when the user manually turns the flow meter without the app or if the concentrator attachment accidentally powers off.

When the motor drive gear rotates, it adjusts the flow rate by rotating the flow meter knob via the flow meter gear and sends the current flow rate to the app by rotating the potentiometer via the potentiometer gear. During the calibration process, the potentiometer values that are associated with 0, 1, 2, 3, 4, and 5 LPM are recorded by the microcontroller and sent to the app. The potentiometer values associated with each 0.5 LPM increment are then interpolated. When a target flow rate is selected on the app, the app sends the previously stored potentiometer value to the microcontroller in



**Figure 4** Full concentrator attachment view with electronics plugged into the 110V wall outlet.

the concentrator attachment. The microcontroller then signals the 12V DC stepper motor to rotate the flowmeter knob until the potentiometer value matches the target value.

## Mobile Application

Mobile applications were developed for iOS and Android devices using Swift and Java, respectively, to allow users to control the oxygen flow using their smartphone (Figure 5). The app communicates with the Bluno Beetle microcontroller using Bluetooth 4.0 (BLE).

The app first requires the user to connect to the oxygen concentrator attachment via Bluetooth 4.0 (BLE) (Figure 6, Step 1 and 2). To initially calibrate their device, the user manually turns the flow meter knob to each one integer increment (eg 0, 1, 2, 3, 4, and 5 LPM) and presses a button on the app to confirm the flow rate (Figure 6, Step 3 and 4). The potentiometer values at each of these flow rates are stored in the app. The potentiometer values associated with each 0.5 LPM increment are then interpolated. After the one-time calibration is finished, users can use the slider bar or one of the four editable button presets (Figure 6, Step 5 and 6) to choose their desired oxygen flow rate (Figure 5). The editable button presets were included to allow for easy adjustment to commonly used flow rates such as those prescribed by a physician for the varying activity levels of sleep, rest, and high activity. Each time the desired flow rate is selected, the phone updates the user with the current flow rate and sends the corresponding potentiometer value to the Bluno Beetle BLE microcontroller in the concentrator attachment. The microcontroller then signals the 12V DC stepper motor to rotate the flowmeter knob until the multi-turn potentiometer value matches the value corresponding to the desired flow rate.

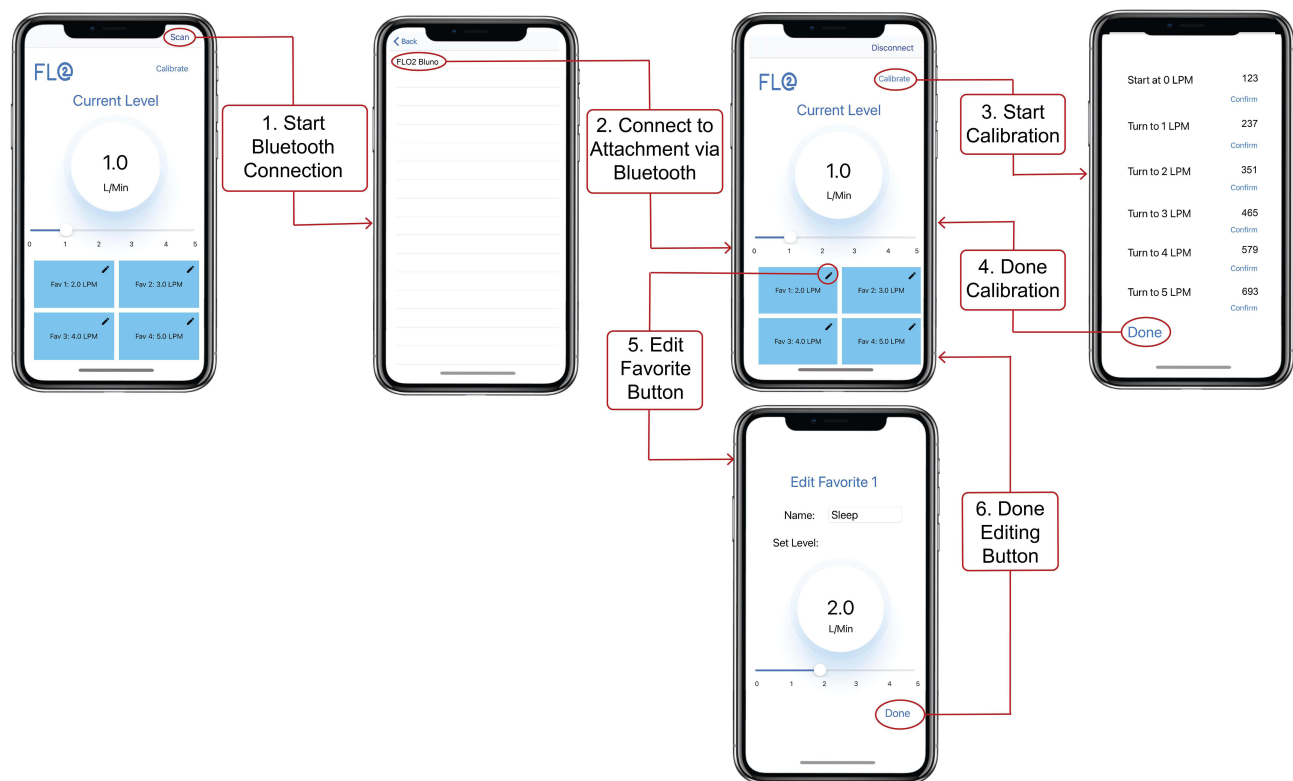
## Results

To evaluate the design, the team tested the prototype according to the following design criteria: Bluetooth range, reliability, flow rate accuracy, and precision (Table 2). Testing was performed on the Invacare Mobilaire V stationary oxygen concentrator.

The Bluetooth range was defined as the maximum distance the app remained connected to the concentrator attachment. A range of at least 30 meters was desirable to ensure that the device would stay connected within the confines of even a relatively large home. The range was tested by identifying how far the app could be from the



**Figure 5** FLO2 application home page with current flow rate display, Bluetooth button, and calibration button. Users can adjust the flow rate by 1) sliding the slider bar to any 0.5 increment value between 0 and 5 LPM or 2) selecting one of the four editable button presets.



**Figure 6** iOS application wireframe displaying Bluetooth connectivity (Steps 1 and 2), calibration (Steps 3 and 4), and editing capability of the four favorite buttons (Steps 5 and 6).

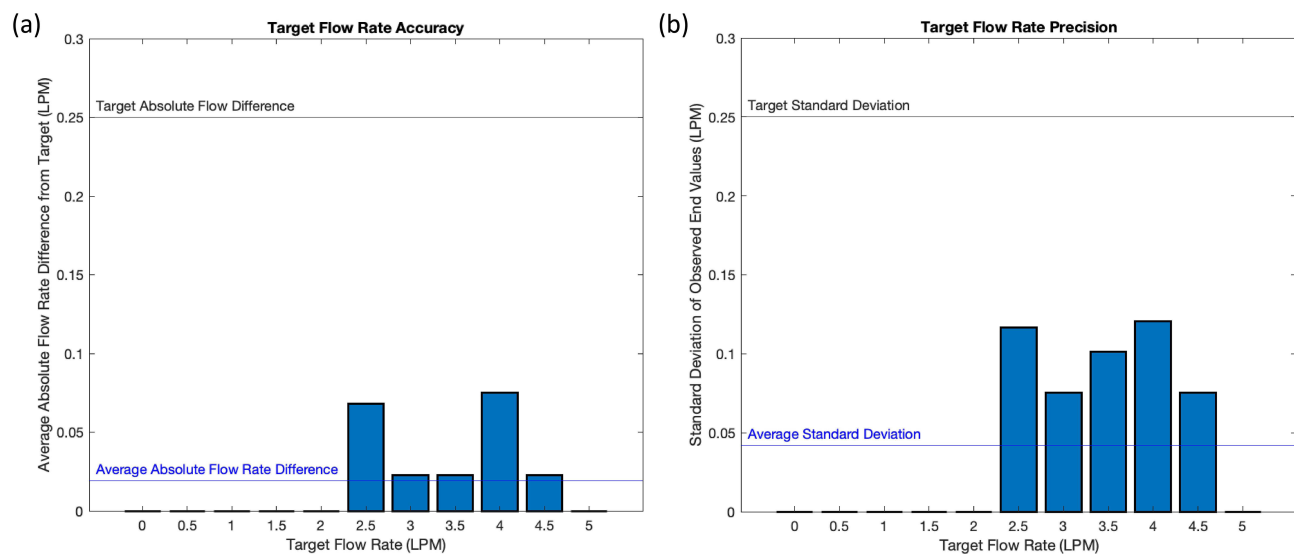
concentrator attachment before the Bluetooth disconnected in an open field. Through testing, it was found that the distance before disconnection was 41 meters in open space, which met the desired goal of  $\geq 30$  meters.

Reliability assessed two different facets of the system: 1) the app's ability to connect via Bluetooth to the concentrator attachment in under 10 seconds and 2) the communication of actions performed on the app to the concentrator attachment. For these reliability standards to be met, at least 195 out of 200 (97.5%) trials performed had to be successful. In testing for reliability, there was successful Bluetooth pairing between the app and the attachment in under 10 seconds for 200 out of 200 (100%) trials. To test for the app adjustment reliability, a button press was

**Table 2** Summary of Prototype Testing results

| Design Criteria     | Performance Criteria                                                                                                                | Results                                                                 |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Bluetooth Range     | Based on 3 trials, the average distance for the Bluetooth connection to disconnect is $\geq 30$ m                                   | Success: Range of 41 m (~140 ft) in open space (n = 3)                  |
| Reliability         | In 195 out of 200 trials, the time needed for successful Bluetooth pairing between the app and Bluno Beetle is < 10 seconds.        | Success: 100% Bluetooth reliability (n = 200)                           |
|                     | In 195 out of 200 trials, user action on the app (button press or slider adjustment) causes the attachment to adjust the flow rate. | Success: 100% App adjustment reliability (n = 200)                      |
| Flow Rate Accuracy  | Based on 122 trials, the average absolute difference between the observed and target flow rate $\leq 0.25$ .                        | Success: Average flow difference from target flow = 0.019 LPM (n = 122) |
| Flow Rate Precision | Based on 122 trials, the average of the standard deviations of the observed flow rates for each target flow rate $\leq 0.25$ .      | Success: Average standard deviation = 0.042 LPM (n = 122)               |

**Abbreviations:** CAD, computer aided design; PLA, polylactic acid; COPD, chronic obstructive pulmonary disease; AC/DC, alternating current/direct current; BLE, Bluetooth low energy.



**Figure 7** Flow rate accuracy (a) and precision (b) across all possible target flow rates (0–5 LPM). The performance criteria (0.25 LPM difference for both flow rate accuracy and precision) are represented by the black lines, while the average results are illustrated by the blue lines in both figures.

administered on the app, and if successful there would be an associated flow rate change. We found that 200 out of 200 (100%) trials were successful, which met the desired standard for app adjustment reliability.

To determine the accuracy and precision of the system, the performance of 122 flow rate adjustments, which included all possible start and end flow rate combinations, was analyzed. Accuracy was accessed by determining the absolute difference between the observed flow rate and the expected flow rate inputted by the user. The overall accuracy was  $\pm 0.019$  LPM, and the average flow difference satisfied the target of 0.25 LPM for all target flow rates (Figure 7a). Precision was accessed by calculating the standard deviation of the observed flow rate for each expected flow rate inputted by the user. The overall precision was  $\pm 0.042$  LPM, and the average standard deviation satisfied the target of 0.25 LPM across all target flow rates (Figure 7b). For both accuracy and precision, a maximum absolute difference or standard deviation of 0.25 LPM was acceptable because supplemental oxygen is prescribed in increments of 0.5 LPM. Therefore, any difference from the desired value of less than 0.25 LPM is highly unlikely to negatively affect the patient. Overall, the data demonstrated that the system can adjust the flow rate of the oxygen concentrator with a high level of accuracy and precision.

## Discussion

With the increasing prevalence of chronic lung diseases and the COVID-19 pandemic, it is essential that home oxygen therapy patients have immediate control over their supplemental oxygen flow. To address this, the FLO2 oxygen concentrator control system was developed to reduce the risks of hypoxia and hyperoxia associated with movement for manual oxygen flow adjustment. For example, many patients turn the flow meter to 4–6 liter/min and leave it there all day due to the difficulty in changing it, which may risk toxic effects when the dose of oxygen is higher than the body's demand. With our system, complications of hyperoxia can be avoided with the ability to turn flow rates down once the patient returns to a resting position. When paired with a pulse oximeter, our system also ensures with prescribed target SpO<sub>2</sub> range (ex. a prescription of 90% to 92% SpO<sub>2</sub>). Increased accessibility may improve therapy compliance, increase patients' quality of life, and lower the risks of adverse blood oxygen levels associated with home oxygen therapy. This two-part control system includes a mobile user application as well as a stationary attachment module to control for a 0–5 liter/min oxygen output flow rate.

Since home oxygen therapy is a medical treatment, the process of home oxygen therapy and any adjustments to the established practice must be accurate, precise, and closely monitored for reliability. Extensive testing of the FLO2 system concluded that the device successfully met the design criteria for precision, accuracy, and reliable



performance. An accuracy of  $\pm 0.019$  LPM proved that the device adjusts the oxygen concentrator to the desired flow rate, which is essential for patients who have exact calculated oxygen prescriptions from their physicians. Similarly, a standard deviation of  $\pm 0.042$  LPM provides evidence of a high level of precision in receiving consistent flow rate values from the oxygen concentrator. Finally, there is reliable pairing of the smartphone app with the attachment module and consistent registration of any inputs by the user. Due to the combination of these experimental successes, patients may confidently rely on the device to provide the desired flow rate without a risk of a failed connection between the app and attachment module. Even in the case of an unforeseen malfunction, the device preserves the natural function of the oxygen concentrator, allowing patients to manually adjust the flow rate whenever necessary.

While the final prototype met all pre-determined design criteria, certain limitations in design and testing allow for future improvements. The maximum distance of Bluetooth connectivity was measured in open space instead of inside a residence, the likely setting for the device. Future testing should take into account various building materials, wall thickness, or size of the residence to accurately determine the Bluetooth range. Additionally, flow rate measurements were read from a floating ball flow meter attached to the concentrator, which is less precise than a digital readout found on more modern oxygen concentrators. To improve the accuracy of flow rate measurements, future trials should measure the actual output from the nozzle of the machine.

While our user needs assessment demonstrated a concern for the lack of remote adjustability and domiciliary accessibility, we had a small survey sample size of 12. To strengthen our user needs assessment, future work will conduct a more extensive survey with a larger, diverse sample of stationary oxygen concentrator users. Further testing of the device will involve evaluating its accuracy and reliability through feedback from oxygen therapy patients, with iterative improvements to be made based on their user experience. Future studies should investigate the compatibility of the FLO2 system with other stationary oxygen concentrator models and evaluate its clinical impact on patients with various respiratory conditions through randomized trials. The device can also be expanded to work with other oxygen therapy devices such as portable oxygen concentrators or compressed gas tanks. Moreover, we can integrate pulse oximetry with this oxygen control system to automate flow rate adjustment based on a patient's blood-oxygen concentration. This system could also be adapted to control hospital oxygen, improving the accessibility of oxygen control for both patients and doctors in the future.

## Conclusion

We developed and tested a novel oxygen control system to wirelessly control the flow rate of supplemental oxygen during home oxygen therapy. As an external add-on to existing oxygen concentrators, this system was demonstrated to be an accurate, precise, and reliable method for oxygen therapy patients to wirelessly adjust their oxygen flow rate within 41 meters. By eliminating the risks of hyperoxia and hypoxia associated with physical adjustment of oxygen concentrator flow meters, our system offers a safer and more accessible method for patients to control oxygen flow. Future work should focus on validating the accuracy and reliability of our device in a clinical study, exploring compatibility with other oxygen therapy devices, and integrating pulse oximetry to automate flow rate adjustment based on a patient's blood-oxygen concentration. Further research and development of this device can lead to its integration with current oxygen therapy practices, ultimately improving oxygen flow control accessibility for both patients and healthcare providers.

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## Disclosure

The authors report no conflicts of interest in this work.

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